MPBI01 – EXERCISES (ENG)

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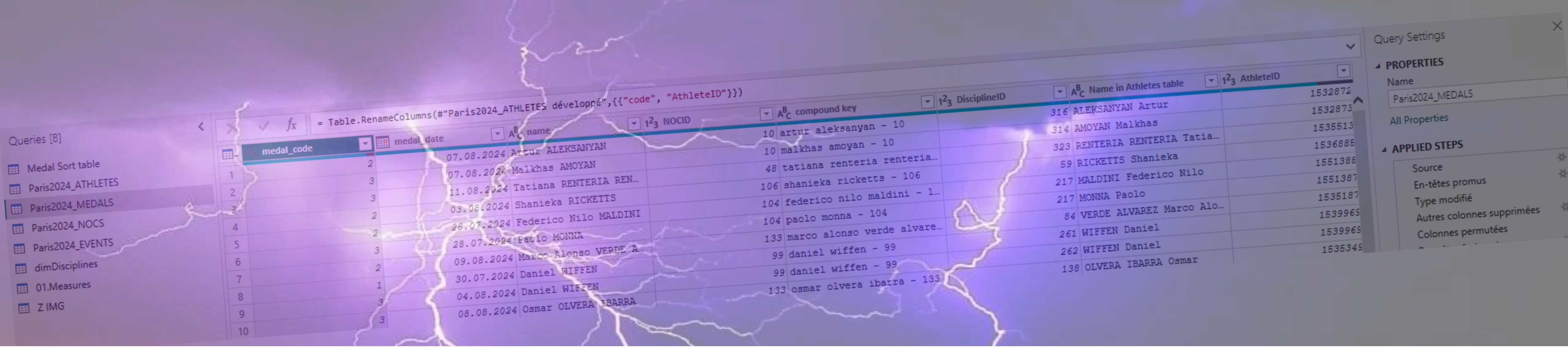


CREATING THE DATA MODEL

Data sources connections

Connect to a data source: Home > Get data

1. Combine all rows (766) of the files contained in the ***Commandes-Orders folder*** into a FactOrders query.
2. The Swicrosoft_prod_prix file contains both tables tbl_prodcat and tbl_prod. Merge them with the [Number] field. Rename to dimProducts. Remove the option to load the 2 original tables so as not to load them into the data model and move them to a 'Noload' group (to be created in the Power Query panel).
3. Create a new Access query and select the ***SwiCrosoft_Données.accdb*** database. Select and load the two Categories and Subcategories tables. Rename to dimCat and dimSouscat
4. Add the customers table from the CSV file ***SwiCrosoft_CLIENTS_master.csv***. Rename to dimClients. CAUTION: Select the semicolon in the column separator and click on the 'Advanced' button to select UTF-8 in the encoding if not detected by Power BI.
5. Add the dimCountriesISO table from the PDF file ***CountriesISO3166.pdf***. Choose a single page from the list and choose 'transform' then filter the list of the 9 tables included in the document and extract the Data column then promote the headers and detect the type.
6. Add a query on the JSON file ***'CountriesContinentsJSON.json***'. Convert to table and expand column 1 keeping only [name],[cca2], [cca3], [region], and [subregion]. Expand [name] with only [common]. Disable loading and move to the 'Noload' group.
7. Merge dimCountriesISO with CountriesContinentsJSON and add [region] and [subregion] to the country table. Detect types and then exit Power Query (Home> Close & Apply).

Files available here: https://github.com/artofnet-training/MPBI01/tree/main/SWICROSOFT

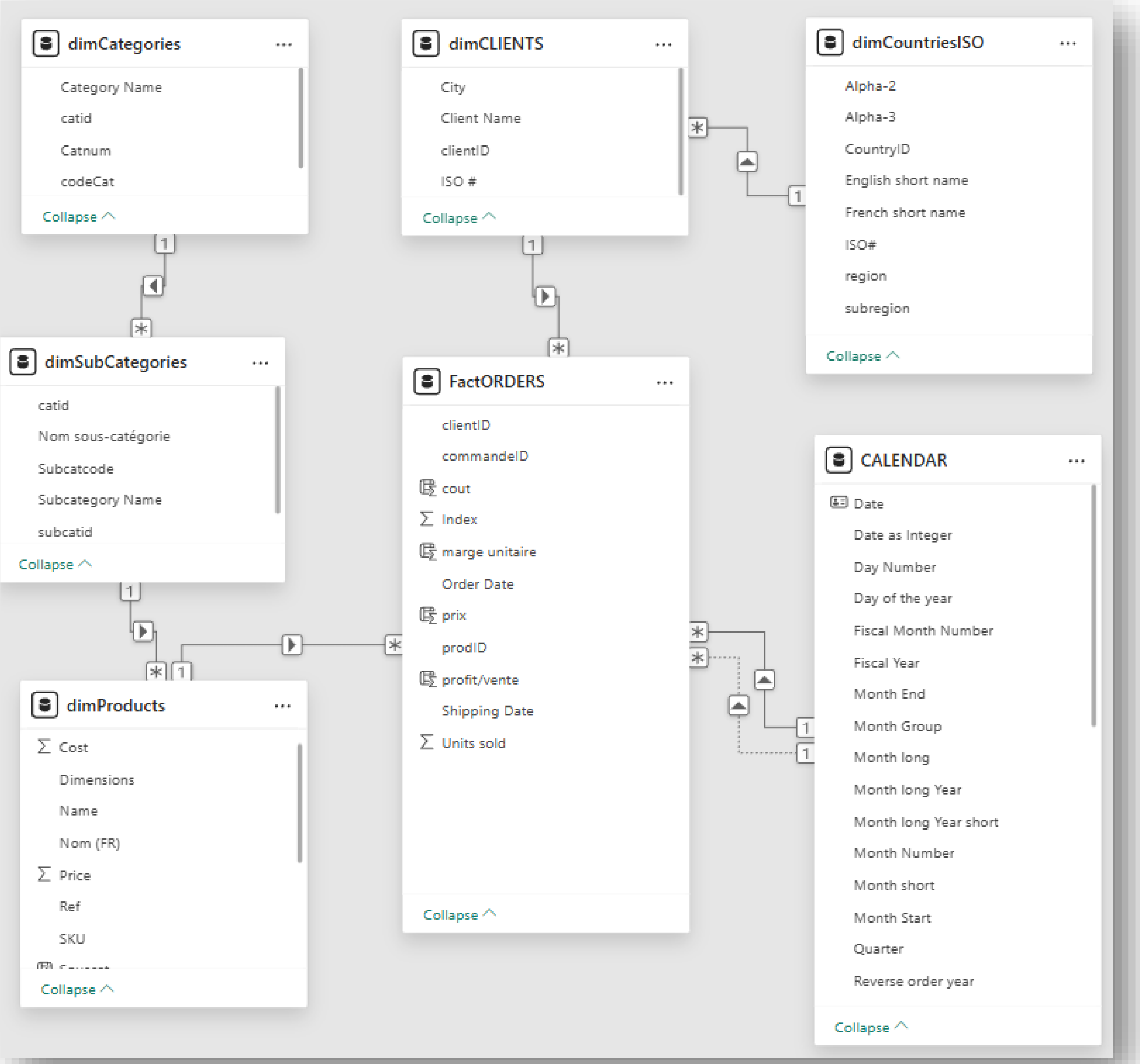
Generate the model view

Navigate to the Model view mode in Power BI

1. In the model view, create the necessary relationships between tables.
2. Add a Calendar table by inserting the DAX script provided for this purpose. Choose the start and end year.

   Mark as date table and select the date field in the calendar table as the date reference.
3. Create a manual table (Home > Enter Data) for storing measures. Rename '01 Measures'.
4. Create a 'hr_cat' hierarchy in the dimProducts table and add [Cat], [Subcat] and [Ref].
5. Create a ‘hr_year' hierarchy in the Calendar table and add [Year], [Quarter], [Month Short] and [Date].

Multi-source data model



Create calculated columns

1. Add two columns to display the cats and subcats in the dimProducts table:

   Cat = RELATED(dimCat[Category Name])

   Subcat = RELATED(dimSubCat[Subcategory Name])

2. Hide the cat and subcat tables from the diagram mode.

3. Add two calculated columns to the factOrders table that evaluates the total per order line and then the profit per sale:

   Total/Sale = RELATED(dimProducts[Price(CHF)])* factOrders[NB Sale]

   Profit/Sale = (RELATED(dimProducts[Price(CHF)])-RELATED(dimProducts[Cost(CHF)]))* factOrders[NB Sale]

Create a Measure Table

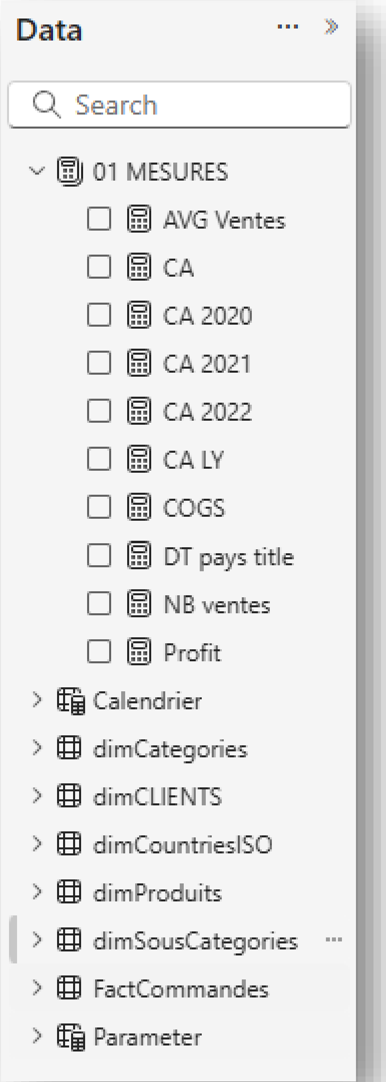
1. Add the following measures:

   Revenue = SUMX(FactCommandes,FactCommandes[NB vente]*RELATED(dimProduits[Price]

   Profit = SUM(FactVentes[Profit])

   COGS = SUMX(FactCommandes,FactCommandes[NB vente]*FactCommandes[cost])

   NB sales = COUNTROWS(FactCommandes)



Create visuals in Power BI Desktop

Create a matrix visual

1. Insert a matrix visual that displays [Revenue] in the "Values" well and [hr_cat] in "Lines".
2. Sort by [Revenue] descending. Using the ellipsis icon at the top right of the visual.
3. Expand and collapse [hr_cat] by using the arrows in the header icons or by right-clicking.
4. Apply conditional formatting (icons) to [Revenue].
5. Insert a segment on dimCategories[Category Name].
6. Add a title and logo.
7. Add a scrim in the background. Download the scrim library from github.com.
8. Rename the page 'Cat Hierarchy’.
9. Add a filter on all pages limiting the possible values of Calendar[Year] to 2020,2021 and 2022.

The 'Cat Hierarchy' page

SWICROSOFT

Sales by categories, subcategories and products

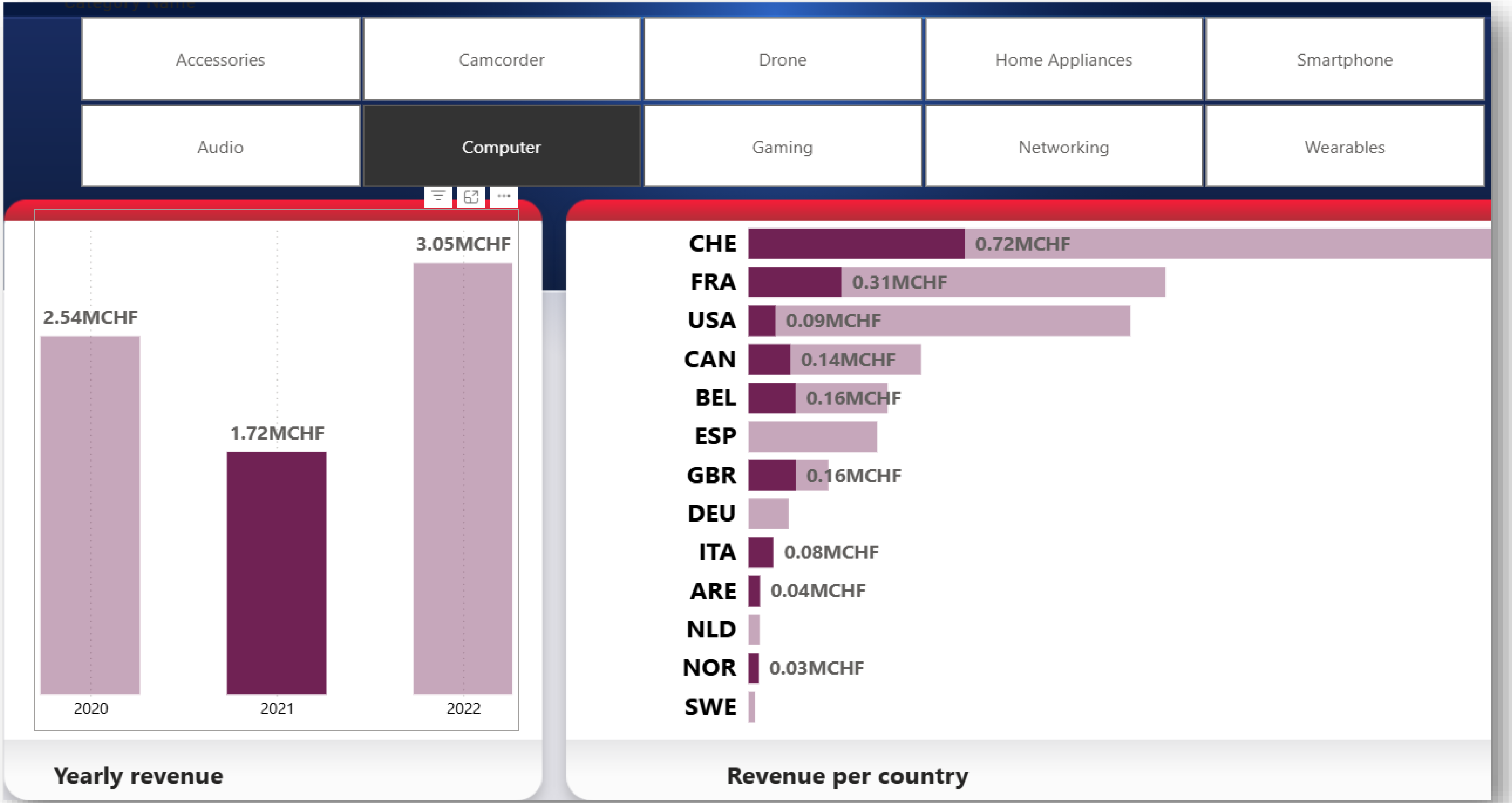
- Accessories
- Audio
- Camcorder
- Computer
- Drone
- Gaming
- Home Appliances
- Networking
- Smartphone
- Wearables

Cat	Revenue
Home Appliances	**8'926'451 CHF**
Accessories	**7'715'752 CHF**
Networking	**7'431'413 CHF**
Smartphone	**7'362'838 CHF**
Computer	**7'311'023 CHF**
Mini PC	2'319'817 CHF
Tablet	2'155'596 CHF
Laptop	1'567'650 CHF
Desktop	828'246 CHF
All-in-One	439'714 CHF
Camcorder	**6'790'551 CHF**
Wearables	**6'297'579 CHF**
Drone	**6'066'240 CHF**
Audio	**5'741'527 CHF**
Gaming	**4'121'711 CHF**
Total	**67'765'085 CHF**

Create a Columns + Bars visual

1. Insert a column visual with [Revenue] in "Y-Axis" and [Year] in "X-Axis".
2. Insert a segment on dimCategories[Category Name].
3. Add a horizontal bar visual with [Country] in "Y-Axis" and [Revenue] in "X-Axis".
4. Add the metrics [COGS], [Profit], and [NB sales] in the tooltips of the 2 visuals.
5. Add a scrim in the background and insert two titles below the visuals
6. Rename the page ‘Revenue by Year/Country'.
7. Test Cross-Highlighting and Cross-filtering between the two visuals.

The page 'Revenue by Year / Country'



Create a Combo Visual

1. Create the 3 DAX measures that calculate the [Revenue] for 2020, 2021 and 2022:
   Revenue 2020 = CALCULATE([Revenue],Calendar[Year]=2020)

2. Create the DAX measure that calculates the previous year's [Revenue].

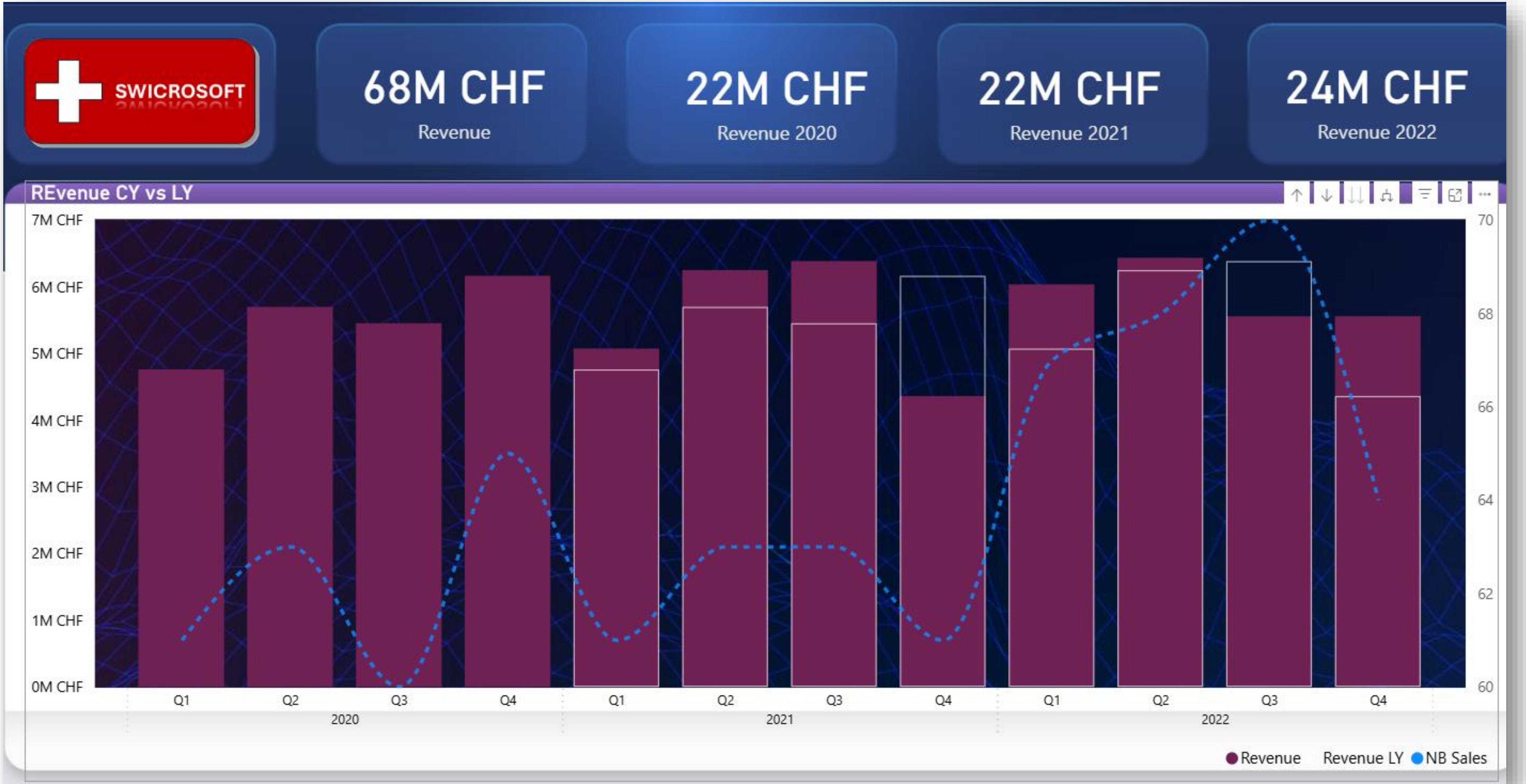
   Revenue LY = CALCULATE([Revenue],SAMEPERIODLASTYEAR(Calendar[Date]))

3. Insérer des visuels de type Carte affichant [Revenue], [Revenue LY], [Revenue 2020] , [Revenue 2021] et [Revenue 2022]

4. Insert a Combo visual with [Revenue] and [Revenue LY] in "Y-Axis of the column" and the year hierarchy in "X-Axis". Then add [NB Sales] in "Line Y-axis" and format the visual as shown on the next page.

5. Use the arrow icons to navigate through the different hierarchical levels of the visual.

6. Rename the page 'CY, LY and NB Sales'.


SWICROSOFT
68M CHF
Revenue
22M CHF
Revenue 2020
22M CHF
Revenue 2021
24M CHF
Revenue 2022
REvenue CY vs LY
7M CHF
6M CHF
5M CHF
4M CHF
3M CHF
2M CHF
1M CHF
0M CHF
70
68
66
64
62
60
Q1
Q2
Q3
Q4
2020
2021
2022
Revenue
Revenue LY
NB Sales

Create a visual with Small Multiples

1. Insert a clustered column visual with [Year] in "X-axis" and [Revenue] in "Y-axis".

2. Drag [Category Name] into "Small Multiples".

3. Add two reference lines (Format > Visual Object > Reference Lines), one for the maximum sales and the other for the average (shade the area below the line for the average).
   Display the data labels for the 2 rows with the Data label style to 'both'.

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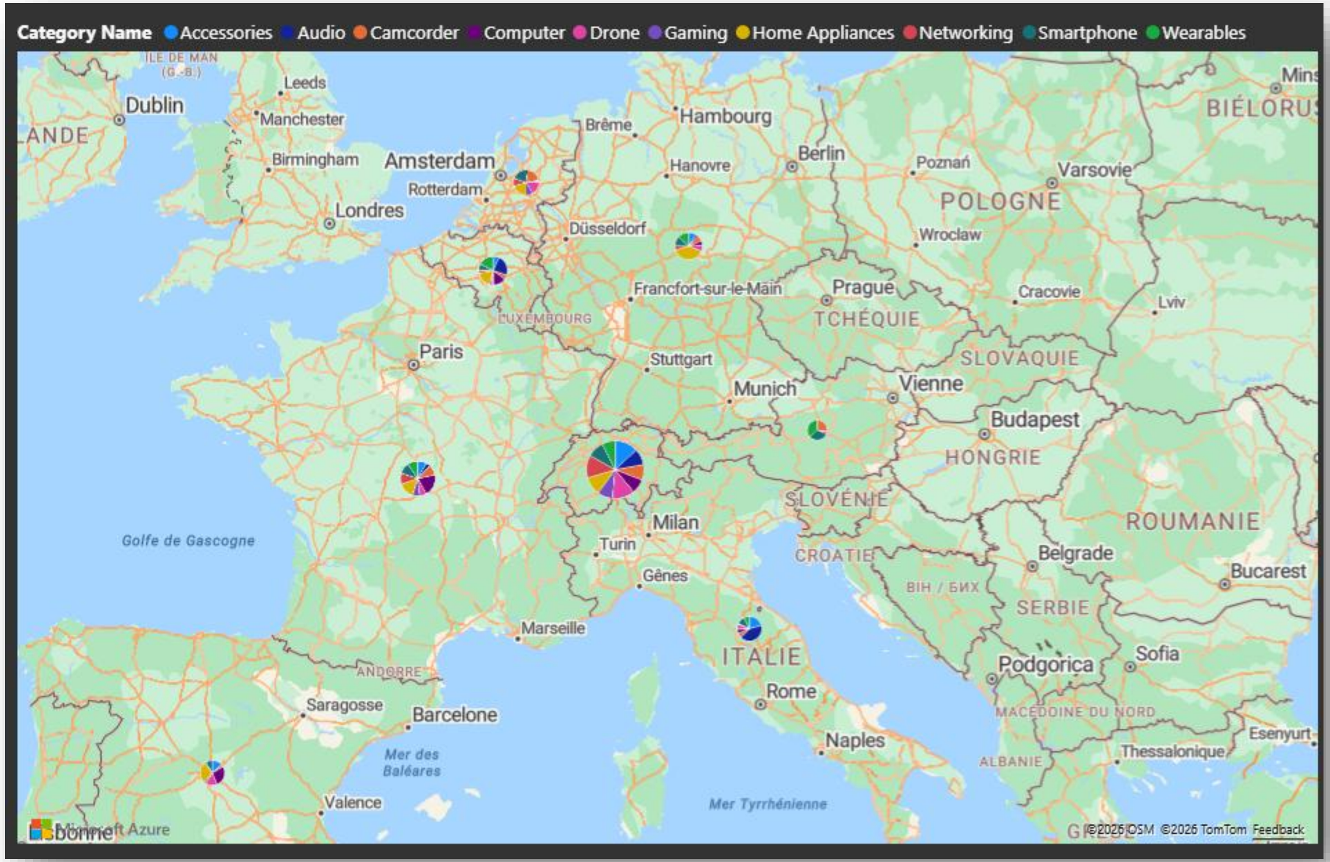
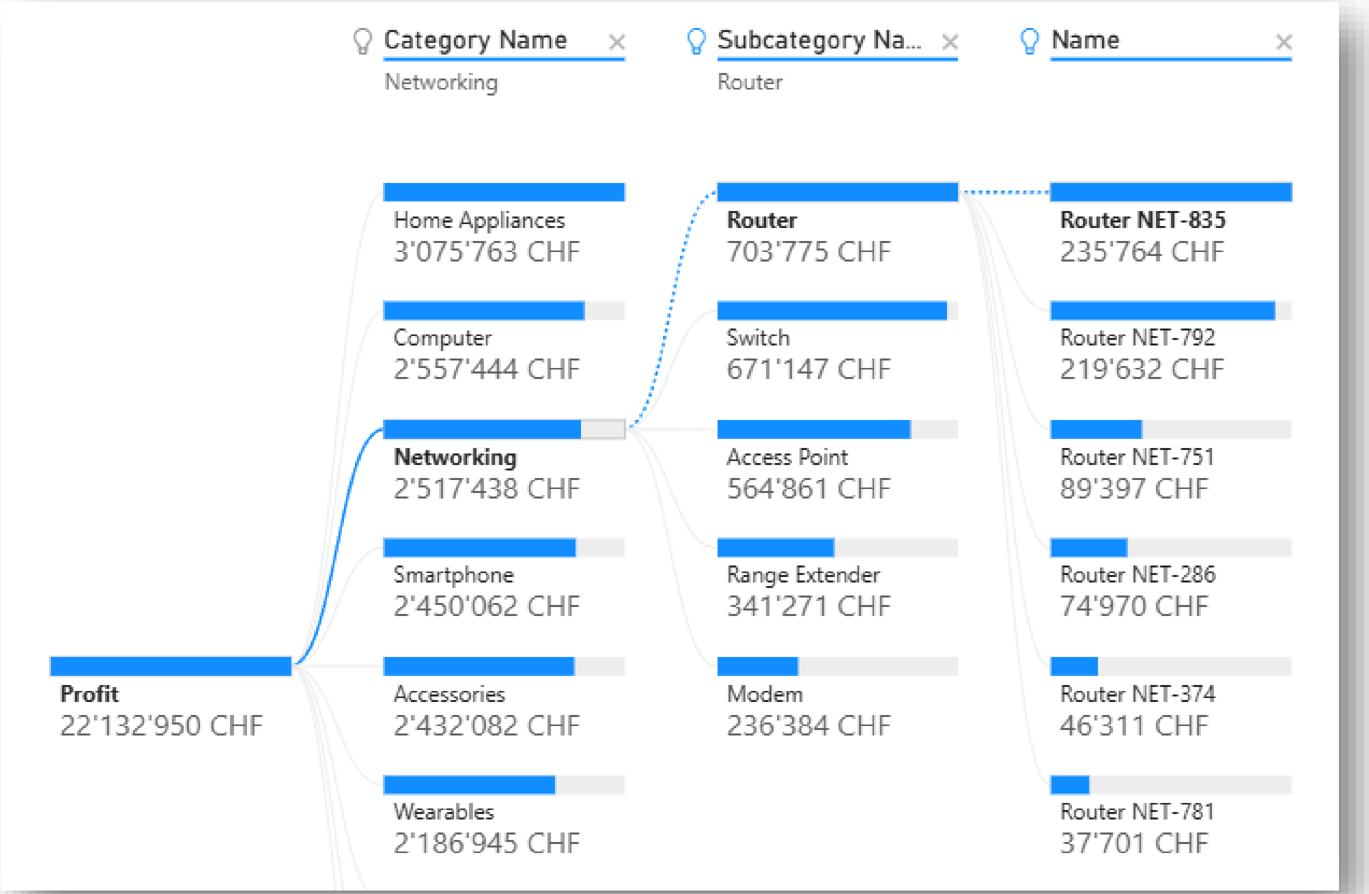
Create an azure map visual

1. Insert an 'Azure Maps' visual and drag dimCountriesISO[English short name] into "Location". Countries present in the model are now represented by bubbles on the map.
2. Add [Revenue] to the "Size" well. The size of the bubbles depends on the revenue value for that country.
3. Add dimCategories[Category Name] in "Legend" to replace the bubbles with pie charts showing the distribution of [Revenue] between the various categories.



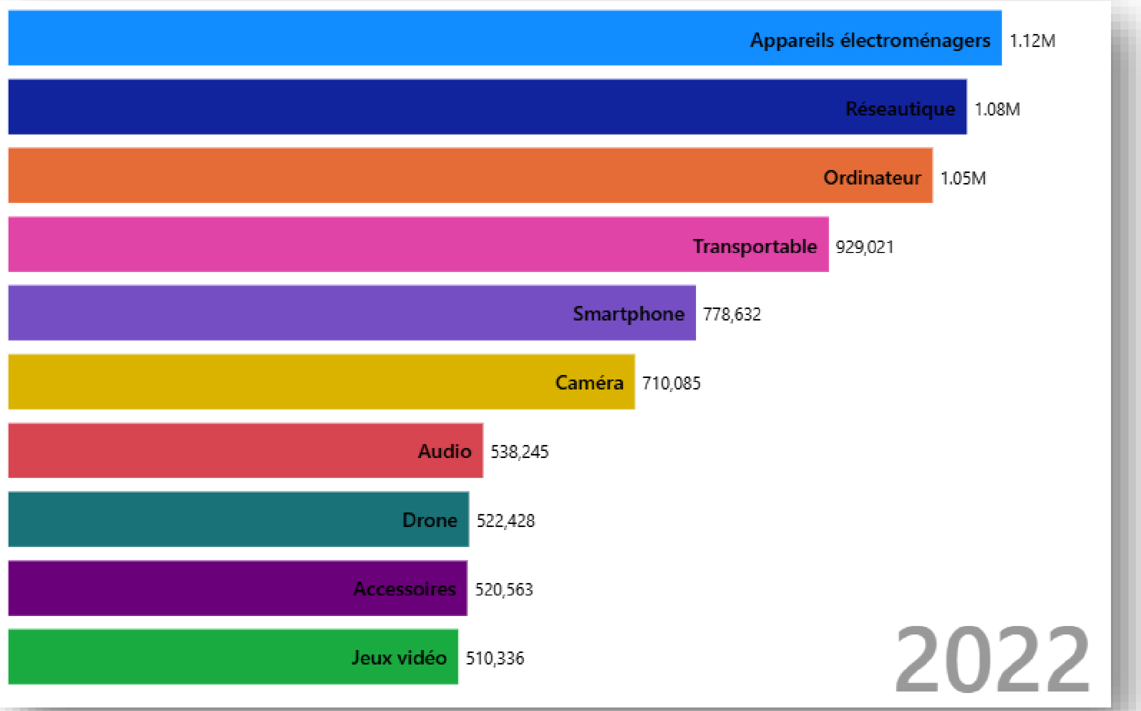
Analysis with a decomposition tree

- Insert a "decomposition tree" visual and drag FactCommands[Profit] into the "Analyze" well and dimCategories[Category Name], dimCategories[Subcategory Name] and dimCountriesISO[English short name] into 'Explain by'.



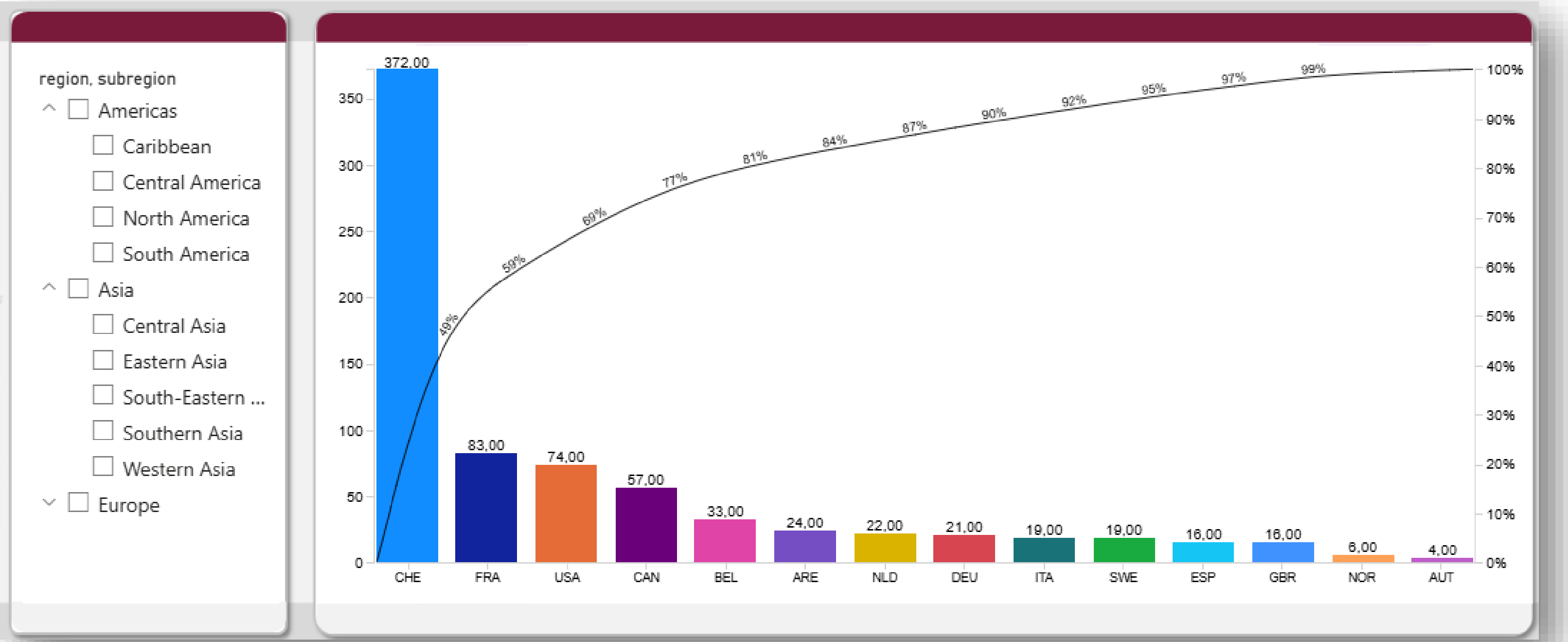
Insert an Animated Bar chart race visual

- Click on the 'Get more visuals' button and in the search bar, type 'animated bar chart race'. Choose the visual created by Wishyoulization and add it to Power BI.
- Insert this visual in a new page and drag Calendar[Year] into the "Period" well, FactOrders[Profit] into "Value" and dimProducts[Category Name] into "Name".
- Check the options in the Format panel, including "Duration" and "TopN"



Insert a Pareto chart

- Click on the 'Get more visuals' button and in the search bar, type 'Pareto'. Choose the visual created by sio2Graphs and add it to Power BI.
- Insert this visual in a new page and drag dimCountriesISO[Alpha-3] into the "Categories" well and [NB Sales] into "Values".
- Insert a slicer and drag dimcountriesISO[region] and dimcountriesISO[subregion] into the "Field" well. Keep only the 'Europe', 'Asia' and 'North America' regions in the visual filter.
- Rename the page 'Pareto' and insert a scrim.



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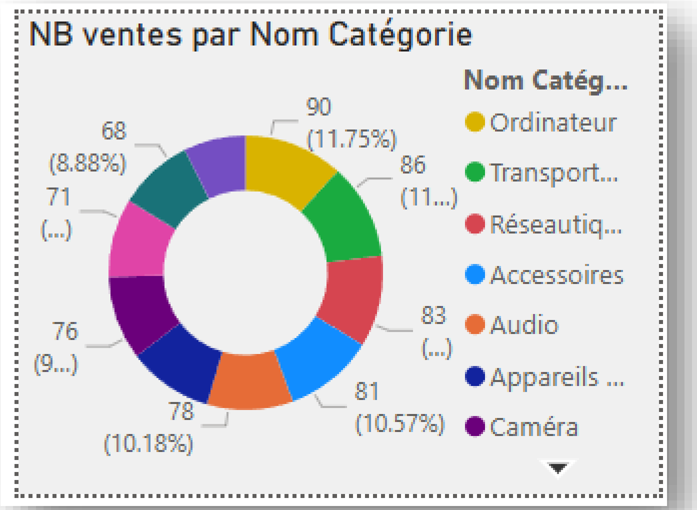
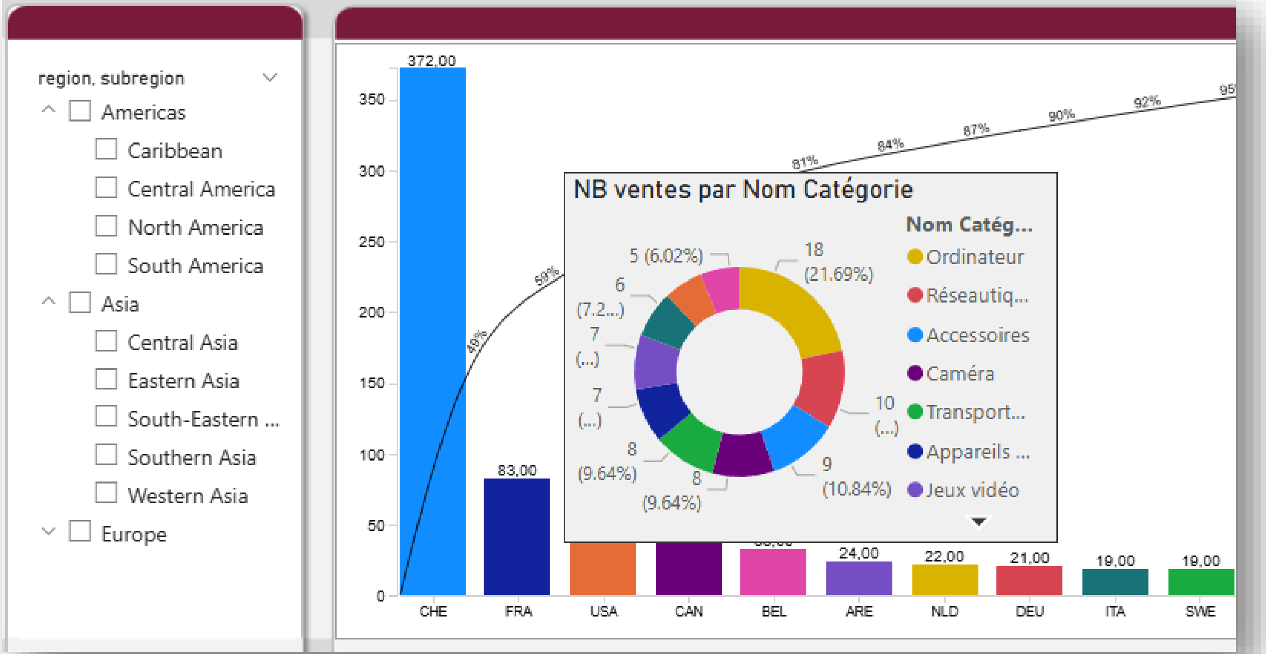
Insert a tooltip page

- Add a new page 'TT NB sales > cat'. In the page format, choose 'Tooltip' as the page type and drag the dimCountriesISO[Alpha-3] field into the 'Show tooltip on' well.
- Insert a donut visual with [NB Sales] in the "Values" well and dimCategories[Category Name] in "Legend". In Format > Size & Style > Background, choose a light color.
- Go back to the Pareto visual and check the tooltip window appearing on the country columns.





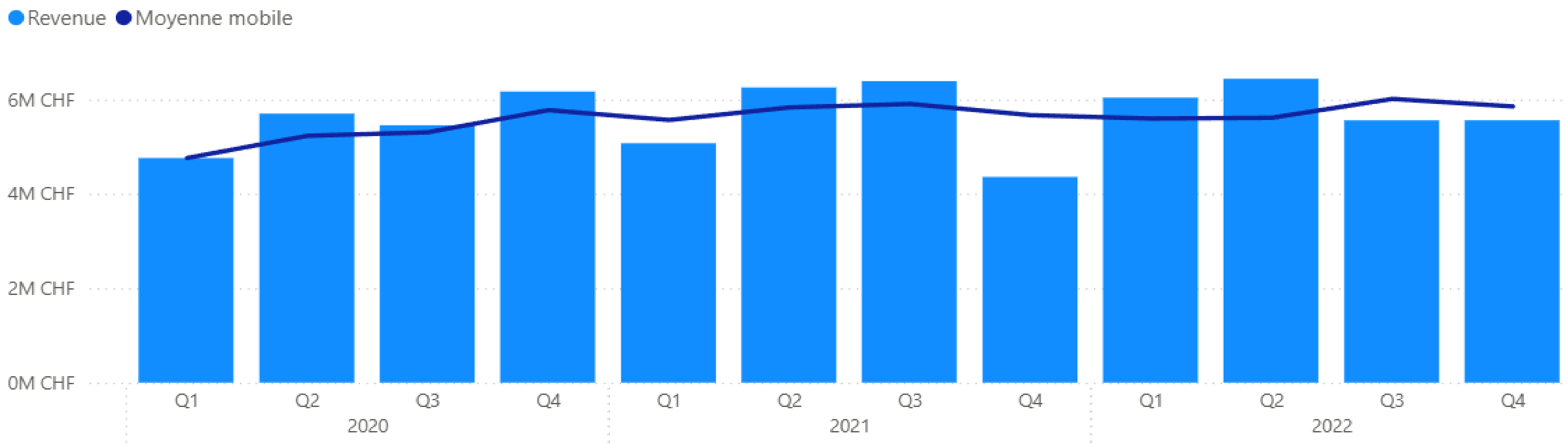
Use quick measurements and visual calculation

- In a new 'Quick/Visual' page, insert a table visual with Calendar[Year Quarter] in 'Rows' and [Revenue] in 'Values'.
- Click on Home > Quick Measure and select 'Year-to-date', drag [Revenue] into 'Base Value' and Calendar[Date] into 'Date'. A new DAX measure (quick measure) is created, add it to the table visual.
- Create another quick measure by choosing 'Cumulative Result', drag [Revenue] into 'Base Value' and Calendar[Date] into 'Date'. Add the new measure to the table visual.
- Click Home > New Visual Calculation. Click on the fx button and select 'Parent Percentage'. In the formula, replace [Field] with [Revenue] and [Axis] with [Quarter Year]. A visual calculation is added to the table. To modify the latter, you have to click the pencil icon to the right of the name in the "Values" well.
- In order to display the percentage format, modify the formula for the visual calculation as follows: Parent Percentage = FORMAT(DIVIDE([Revenue], COLLAPSE([Revenue], [Year Quarter])),"0.0%")
- Insert a Combo visual and drag Calendar[Year Quarter] into "X-Axis", [Revenue] into "Y-Axis of Column". Sort ascending on [Year Quarter].
- Add a new 'Moving Average' visual calculation. Replace [Field] with [Revenue] and Windowsize with 3 to average over 3 periods. Once the visual calculation is created, drag it into "Line Y-axis."

Mesures rapides et calculs visuels

Year/Quarter	Revenue	Revenue %	Running total YTD	Running total on REVENUE
2020/Q1	4'762'756 CHF	7.0%	4'762'756 CHF	4'762'756 CHF
2020/Q2	5'701'358 CHF	8.4%	10'464'114 CHF	10'464'114 CHF
2020/Q3	5'455'708 CHF	8.1%	15'919'822 CHF	15'919'822 CHF
2020/Q4	6'167'397 CHF	9.1%	22'087'219 CHF	22'087'219 CHF
2021/Q1	5'075'765 CHF	7.5%	5'075'765 CHF	27'162'984 CHF
2021/Q2	6'253'177 CHF	9.2%	11'328'942 CHF	33'416'161 CHF
2021/Q3	6'388'147 CHF	9.4%	17'717'089 CHF	39'804'308 CHF
2021/Q4	4'363'558 CHF	6.4%	22'080'647 CHF	44'167'866 CHF
2022/Q1	6'038'581 CHF	8.9%	6'038'581 CHF	50'206'447 CHF
2022/Q2	6'437'539 CHF	9.5%	12'476'120 CHF	56'643'986 CHF
2022/Q3	5'559'780 CHF	8.2%	18'035'900 CHF	62'203'766 CHF
Total	**67'765'085 CHF**	**100.0%**	**23'597'219 CHF**	**67'765'085 CHF**

Revenue and moving average by Year, Quarter and Month



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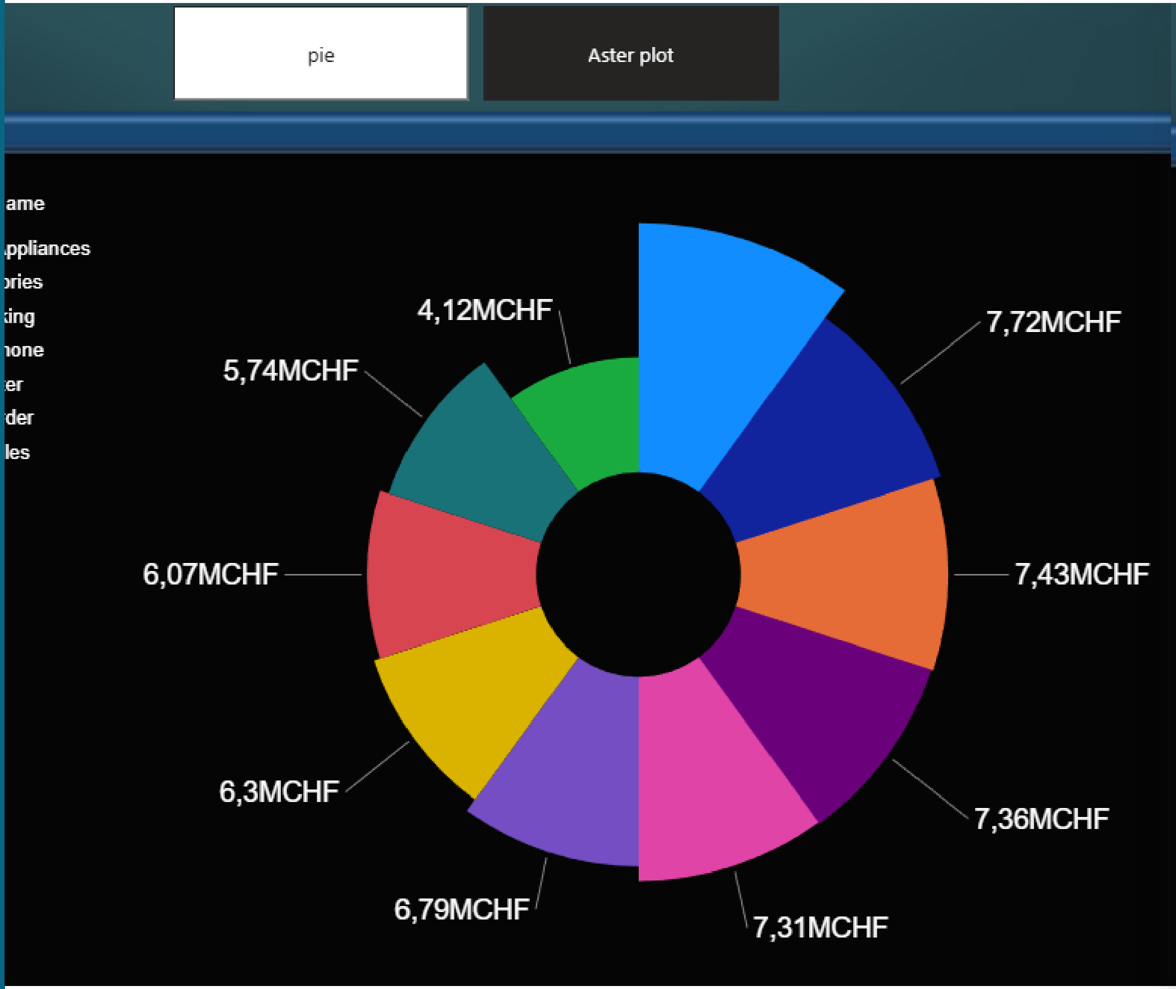
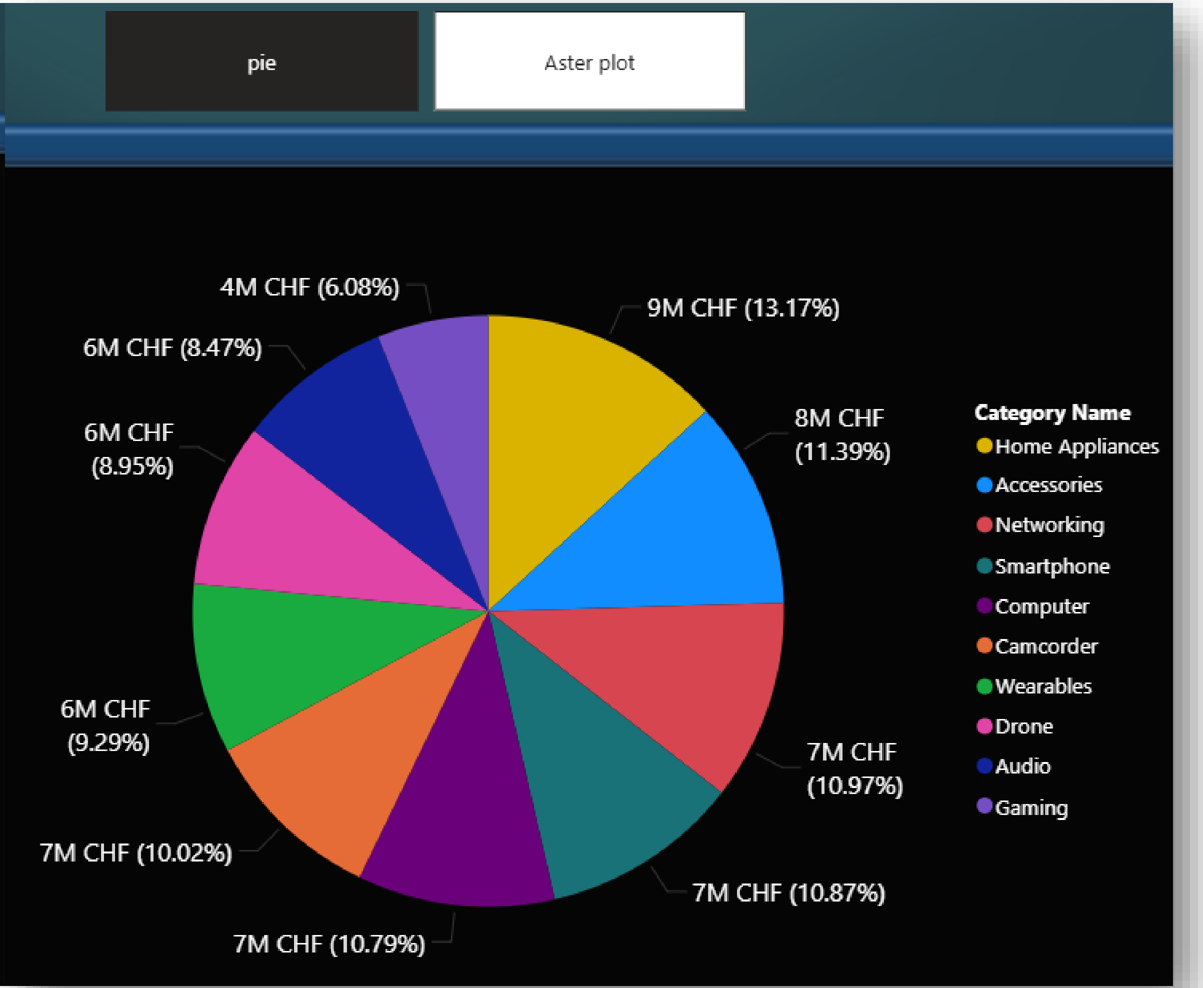
Insert two visuals into one page

- Add a new ‘Pie and Aster by category' page. Add a Pie chart visual with [Revenue] in the "Values" well and dimCategories[Category Name] in "Legend".
- Download a new visual, search for 'Aster plot' from Microsoft and add it to Power BI. Drag [Revenue] into the "Y-Axis" well and dimCategories[Category Name] into "Category".
- Display the Bookmarks pane (View > Bookmarks). Open the selection pane (Show > Selection) and hide the aster plot visual. Center the pie chart visual and save the bookmark in this state (Bookmarks pane> Add). Rename it ‘Pie’.
- In the selection pane, hide the pie chart visual and show the Aster plot visual. Add a new bookmark in this page state. Rename it 'Aster plot'.
- In the Bookmarks pane, right-click on one of the bookmarks and select 'Group'. Drag the 2nd bookmark into the newly created group. Rename the group 'Aster-pie'.
- On the report page, Insert > Buttons > Navigator > Bookmark Navigator.
- Select the bookmark Navigator and go to Format > Bookmarks and choose the 'Aster-pie’ group created earlier from the list of available bookmarks.
- In Power BI Desktop, you need to press Ctrl+click to click on any button such as the bookmark navigator buttons.

Alternate between two visuals with bookmarks





Insert three Top10 visuals with bookmarks

- Add a new 'TOP10' page. Add a funnel visual with [Profit] in the "Values" well and dimProducts[Name (EN)] in "Category". In the Filters pane, go to Filters on this visual and choose 'Top N' filter type for thee[Name (EN)] and then select 'Top' and '10' in 'Show Items'. Then drag [Profit] into "By Value". Click on Apply Filter. The funnel is now limited to the 10 highest profits per product name.
- Insert a matrix visual with dimProducts[Name (EN)] in "Lines" and [Revenue 2020], [Revenue 2021] and [Revenue 2022] in the "Values" well.
- Insert another matrix with dimProducts[Name (EN)] in "Lines" and [Revenue], [Profit] and [NB Sales] in the "Values" well. Apply conditional formatting to the [Profit] column. Sort on [Revenue] descending.
- Display the Bookmarks pane (View > Bookmarks). Open the selection pane (Show > Selection) and hide two out of three visuals and save a bookmark in that state (Bookmarks > Add). Give it an appropriate name.
- Repeat this for the other two visuals to get three separate bookmarks, one for each visual.
- In the Bookmarks pane, right-click on one of the bookmarks and select 'Group'. Drag the other 2 bookmarks into the created group. Rename the group 'TOP 10'.
- On the report page, Insert > Buttons > Navigator > bookmark navigator. In Format > Grid Layout > Orientation, choose 'Vertical' and place the buttons to the left of the visuals. In Format > Bookmarks > Bookmark, choose the 'TOP 10' group from the list of available bookmarks.
- Insert card visuals with the following metrics: [Revenue], [Profit], [COGS], and [NB Sales]. Insert headings to finalize the presentation.

Page with bookmark navigator TOP 10

68M CHF
Revenue

22M CHF
Profit

TOP 10 SALES

Coût global 46M CHF

NB Sales 766

Top10 profit per product

Top10 Rev per prod

TOP 10 per year

Nom (FR)	Revenue 2020	Revenue 2021	Revenue 2022
Barres de son AUD-468	408'510 CHF	184'586 CHF	245'106 CHF
Caméra Web ACC-829	484'632 CHF	20'574 CHF	450'342 CHF
Commutateur NET-639	355'725 CHF	48'195 CHF	656'370 CHF
Compact CAM-673	4'374 CHF	526'338 CHF	333'882 CHF
Drone caméra DRO-714	677'484 CHF		220'320 CHF
Hub domestique HAP-355	406'788 CHF	481'428 CHF	
Milieu de gamme SMR-367	473'755 CHF		383'626 CHF
Tablette COM-262	337'800 CHF	225'200 CHF	867'020 CHF
TV intelligente HAP-888		501'142 CHF	501'142 CHF
TV intelligente HAP-891	192'178 CHF	614'607 CHF	219'373 CHF
Total	**3'341'246 CHF**	**2'602'070 CHF**	**3'877'181 CHF**

Create a parameter table

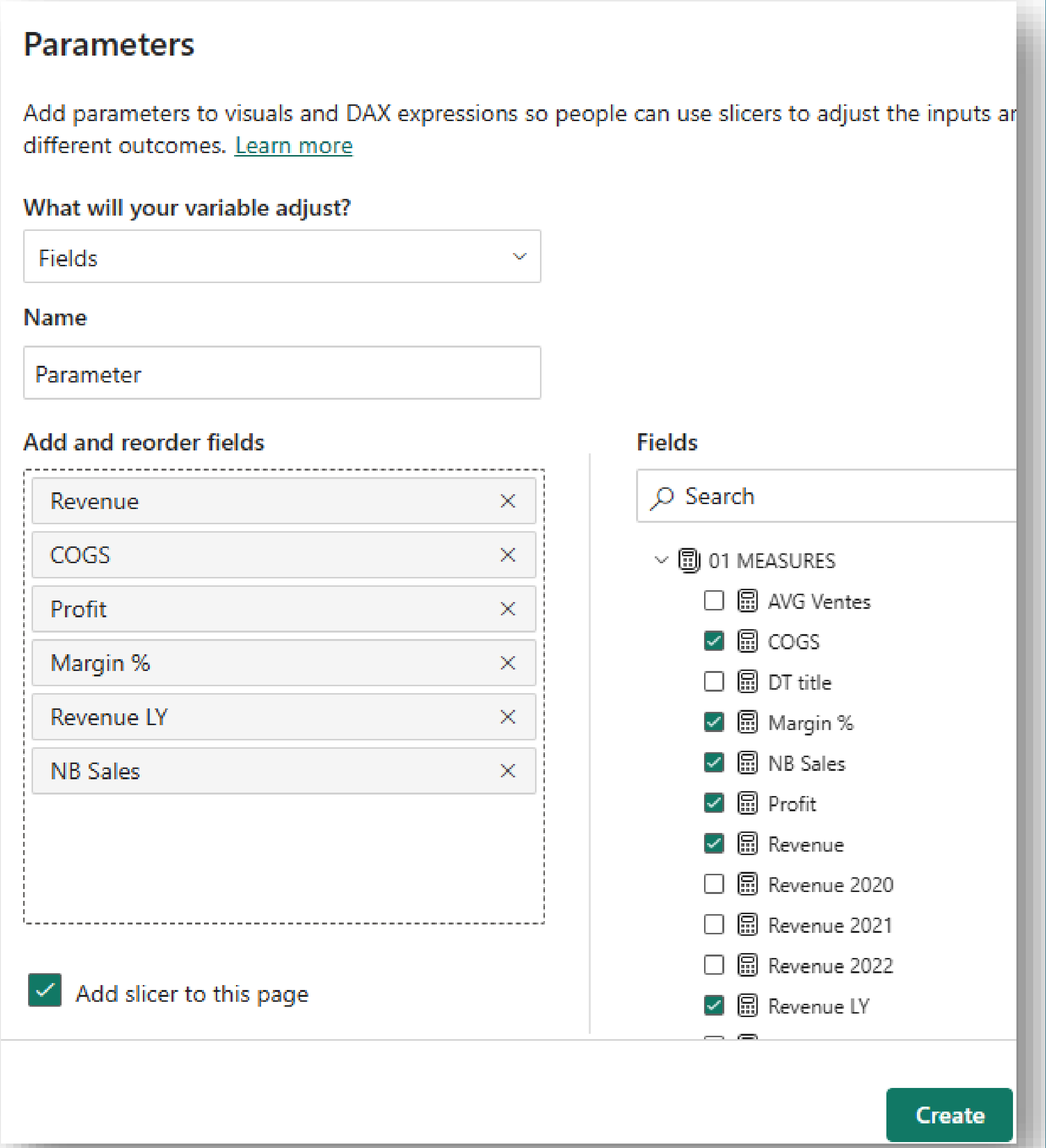
1. Modeling > New Parameter > Fields

   Select the measures in the measures table (they then appear in the list on the left)

   Check the option 'Add slicer...' if you haven't already.

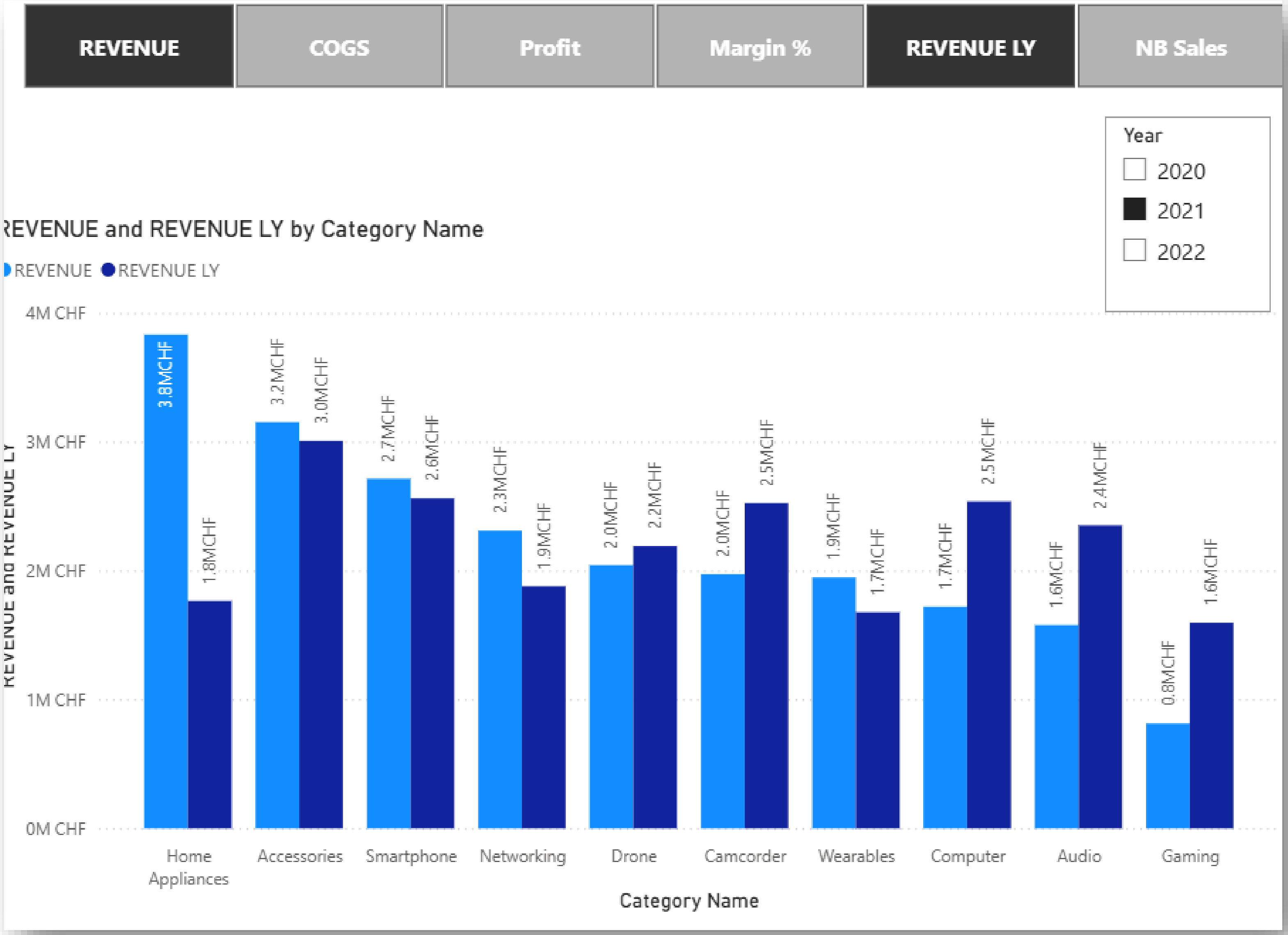
2. A table will be added to the model.

3. A slicer is also added to the page with the names of the measures in the buttons. Change the slicer display type if necessary.



Insert a multi-measures visual

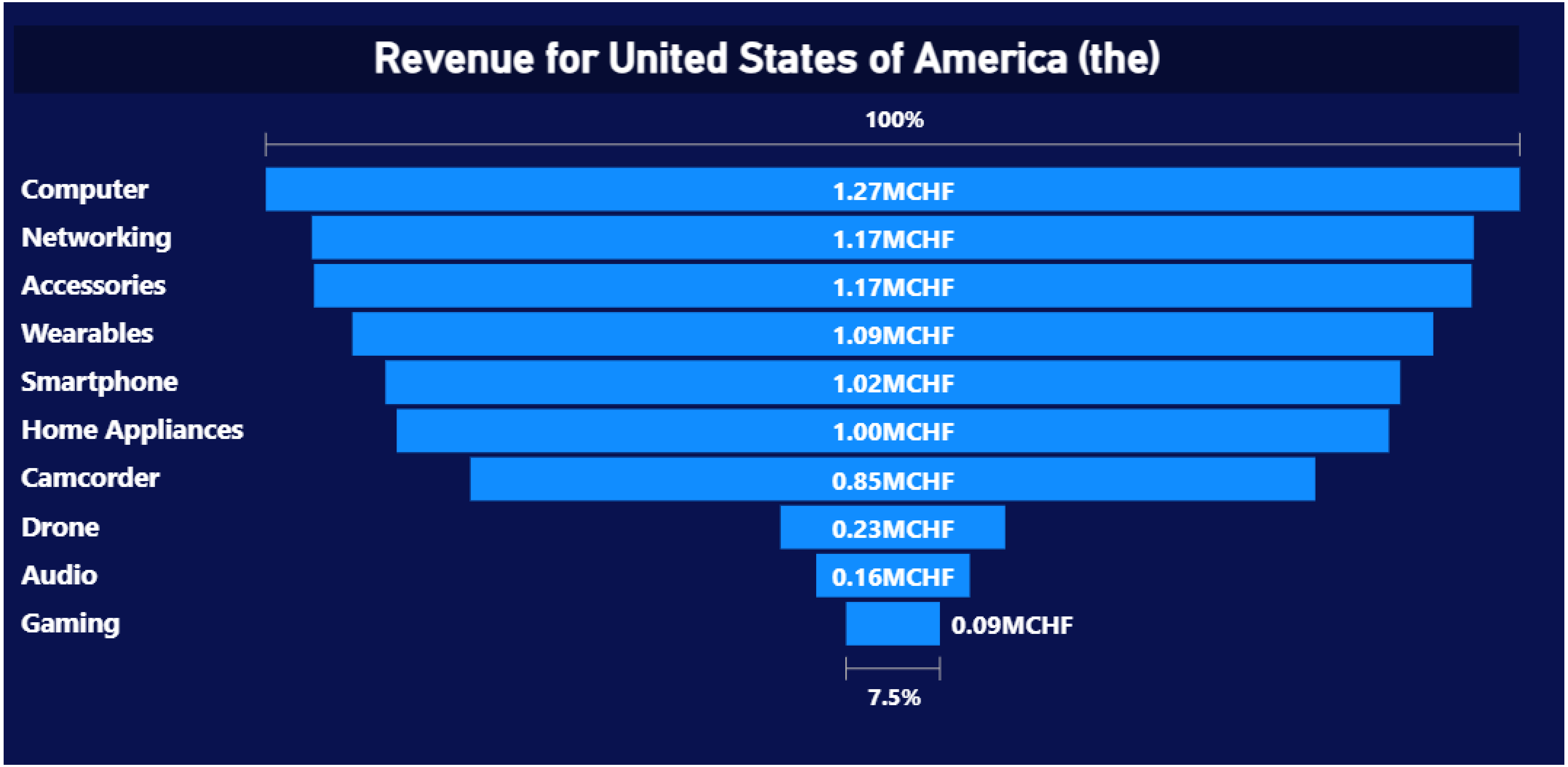
1. Insert a visual in clustered columns with the 'Parameter' field of the table of the same name in the Y axis and the name of the category in the X axis.
2. To compare multiple measures with each other, Ctrl + click on the desired measures in the slicer to display them to the visual.
3. Add a slicer with Calendar[Year] in "Field".



Drillthrough Page over Country

- Create a new page with a funnel visual with [Revenue] in "Values" and dimProducts[Cat] in "Category".
- Go to the Page Format pane and insert:
  - Information on the page > Name: ‘DT Alpha3’.
  - Page Information > Page Type: ‘Drillthrough’
  - Information on the page > Drill through from > dimCountriesISO[Alpha-3]
  - Canvas Background > Color: Choose a dark blue.
- Create a DAX measure displaying a dynamic headline:
  DT title = "Revenue for country "& SELECTEDVALUE(dimCountriesISO[English short name])
- Insert a text box and go to Format > Title > Title > fx and select the DAX measure [DT title] from the drop-down list at the bottom of the window.
- The drillthrough page is now active on any visual displaying [Alpha-3].

The ‘DT Alpha3’ drillthrough page



Revenue drillthrough page

- Create a new page and add a table visual. Drag the following fields into the “Columns” well:
  - dimClients[Client Name] and dimClients[City]
  - [Revenue], [Profit], [Margin %] and [NB Sales]
- Go to the Page Format pane and insert:
  - Page Information > Name: 'DT Revenue’.
  - Page Information > Page Type: ‘Drillthrough'
  - Page Information > Drill through from > [Revenue]
  - Canvas Background> Color > (Choose a color).
- Insert a text slicer with dimClients[Client Name] in “Field”.
- Insert a slicer on Calendar[Year].
- This drillthrough page is now active on any visual displaying [Revenue].

Publish the report to Power BI service

- When the report is ready, click Home > Publish to publish the report and semantic model to the cloud in one of your workspaces (by default ‘my workspace’).
- Open the report in the Power BI service and select visuals to pin to a new dashboard (to be created when prompted).
- Try pinning an entire page to an existing dashboard.
- Add tiles to your dashboard, i.e. the company logo.
- The dashboard can now be shared with your colleagues (if they are using a PRO licence or above).

Knowledge is power

Francis Bacon (1561 - 1626)